

PH784 Advanced Social Epidemiology: Methods

Week 7: Mediation & Review

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Today's Plan

1. Simulating disparities
2. Mediation
 - Basics
 - Applications in social epi
 - Connecting back to 1
3. High-level Review

Let's Step Back: Anatomy of Causal Estimand

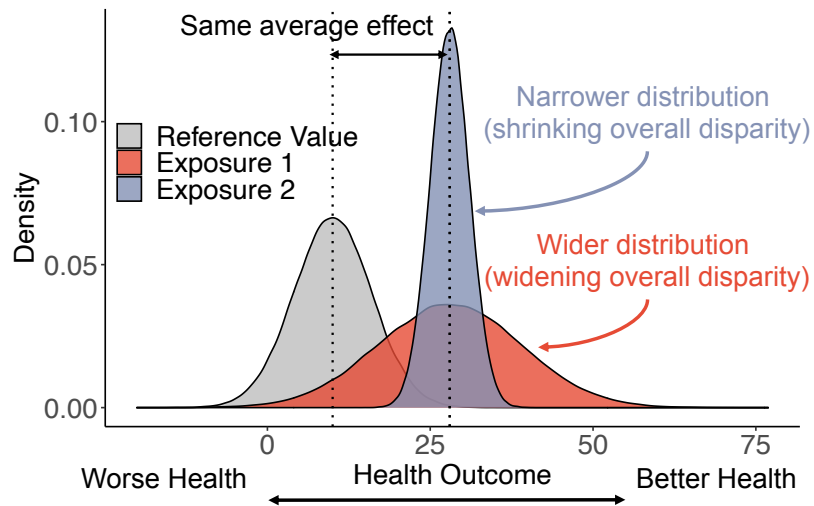
$$E[Y^{a=1}] - E[Y^{a=0}]$$

 **This comparison is defined based on the choice of:**

1. Hypothetical interventions to define potential outcomes
 - Well-defined treatment, directionality, regimen (time-varying, dynamic, stochastic)
2. Study population (e.g., mean among whom?)
 - Subpopulation: ATE vs CATE, ATT, ATC, etc
3. Reference (e.g., $a = 0$)
 - vs natural course
4. **Summary measure** (e.g., mean difference)

Summarizing Changes in Population Health

Distribution beyond mean



- Hypothetical interventions may have:
 - Same “effect” on average population health
 - Different implications for distributions
 - i.e., health disparities

Shifting the definition of “causal effect”

- “How would the hypothetical intervention change health disparities?”
 - e.g., $(E[Y^{A^g}|SES = high] - E[Y^{A^g}|SES = low]) - (E[Y|SES = high] - E[Y|SES = low])$
 - for stochastic intervention A^g
 - “The gap-closing estimand” (Lundberg 2024)
 - Example: Social isolation and social inequalities in all-cause mortality (Lunar et al. 2025)

Population Interventions and Health Disparity

- Interventions may alter health disparities across social groups via:
 - **Socially patterned effect heterogeneity**
 - $E[Y^{a=1} - Y^{a=0} | L = l] \neq E[Y^{a=1} - Y^{a=0} | L = l^*]$
 - Narrow disparities if more beneficial among minoritized groups
 - Widen disparities if less beneficial among minoritized groups
 - **Disproportionate change in exposure distribution**
 - If minoritized groups experienced greater changes in exposure, they benefit more
 - Ignored in the “all treated” scenario of CATE

! Estimating CATEs alone is inadequate

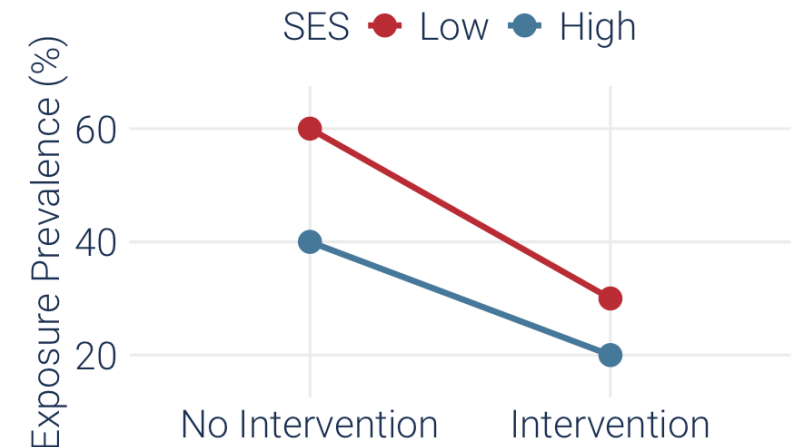
When do interventions result in disproportionate change in exposure?

Population Interventions and Health Disparity

When do interventions disproportionately change exposure distributions?

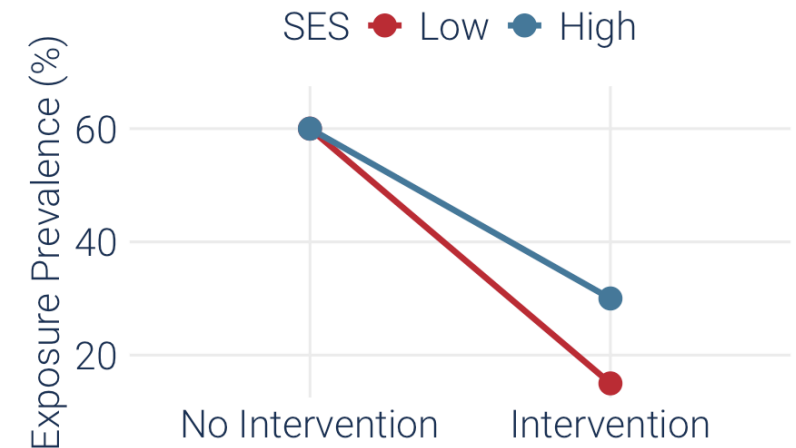
- **Existing exposure disparity**

- More prevalent among the low SES group
- Even the same relative reduction would result in greater absolute reduction
 - e.g., reducing exposure by half
 - 60% to 30% (30 percentage points reduction)
 - 40% to 20% (20 percentage points reduction)



- **Disproportionate reduction in the exposure**

- Even when there is no exposure disparity
 - Exposure prevalence 60% in both SES groups
 - 60% to 15% (quarter)
 - 60% to 30% (half)
 - Distinguish this from effect heterogeneity
- *Example?*
 - Water fluoridation



Population Interventions and Health Disparity

How do we simulate effects on health disparities rather than average health?

- Key ingredients of health disparities
 - **Effect heterogeneity**
 - $E[Y^{a=1} - Y^{a=0} | L = l] \neq E[Y^{a=1} - Y^{a=0} | L = l^*]$
 - **Disproportionate changes in exposure**
 - How do we incorporate disproportionate changes in exposure in causal estimand?
 - **Stochastic intervention**
 - Various estimators (e.g., Muñoz and Van Der Laan 2012)
 - e.g., Standardization (as in Ahern et al. 2016)

Population Interventions and Health Disparity

How do we simulate effects on health disparities rather than average health?

ID	Y	A	L	$\tau(L)$	$a = 1$	$a = 0$	$Y^{a=1}$	$Y^{a=0}$
1	Y_1	1	ℓ_1	τ_1	1	0	Y_1	$Y_1 - \tau_1$
2	Y_2	0	ℓ_2	τ_2	1	0	$Y_2 + \tau_2$	Y_2
3	Y_3	0	ℓ_3	τ_3	1	0	$Y_3 + \tau_3$	Y_3
...	...	...	...	...	...	...	...	...
n	Y_n	A_n	ℓ_n	τ_n	1	0	$Y_n + (1 - A_n)\tau_n$	$Y_n - A_n\tau_n$

- **Simple standardization for static interventions**

- #1 Assign everyone to be treated $a = 1$ (6th column) and untreated $a = 0$ (7th column)
- #2 Predict potential outcome under each scenario ($Y^{a=1}$ & $Y^{a=0}$)
 - Based on estimated effects τ_i
- #3 Calculate mean of $Y^{a=1}$ and mean of $Y^{a=0}$ across all individuals
- #4 Calculate the difference between the two means & bootstrap
 - ATE: $E[Y^{a=1}] - E[Y^{a=0}]$

Population Interventions and Health Disparity

How do we simulate effects on health disparities rather than average health?

ID	Y	A	L	$\tau(L)$	A^g	Y^{A^g}
1	Y_1	1	ℓ_1	τ_1	1	Y_1
2	Y_2	0	ℓ_2	τ_2	0	Y_2
3	Y_3	0	ℓ_3	τ_3	1	$Y_3 + \tau_3$
...	...	...	...	...	...	...
n	Y_n	A_n	ℓ_n	τ_n	A_i^g	$Y_n + (A_i^g - A_n)\tau_n$

- **Standardization for stochastic interventions**

- #1 Assign exposure distributions under a stochastic intervention of interest
 - e.g., randomly removing exposure by 50%
 - e.g., lowering propensity scores by 50%
- #2 Predict potential outcome Y^{A^g} based on estimated effects τ_i
- #3 Calculate mean of Y^{A^g} and mean of observed outcome Y across all individuals
- #4 Calculate the difference between the two means & bootstrap
 - Stochastic intervention effect: $E[Y^{A^g}] - E[Y]$
 - compared to “no intervention”

Population Interventions and Health Disparity

How do we simulate effects on health disparities rather than average health?

ID	Y	A	L	$\tau(L)$	A^g	Y^{A^g}
1	Y_1	1	ℓ_1	τ_1	1	Y_1
2	Y_2	0	ℓ_2	τ_2	0	Y_2
3	Y_3	0	ℓ_3	τ_3	1	$Y_3 + \tau_3$
...	...	...	...	...	...	...
n	Y_n	A_n	ℓ_n	τ_n	A_i^g	$Y_n + (A_i^g - A_n)\tau_n$

- **Standardization for stochastic interventions on health disparities**

- #1 Assign exposure distributions under a stochastic intervention of interest
- #2 Predict potential outcome Y^{A^g} based on estimated effects τ_i
- #3 Calculate **disparities** in Y^{A^g} and **disparities** in observed outcome Y
- #4 Calculate the difference between the two **disparities** & bootstrap
 - e.g., $(E[Y^{A^g}|SES = high] - E[Y^{A^g}|SES = low]) - (E[Y|SES = high] - E[Y|SES = low])$
 - “How would the hypothetical intervention change SES-based health disparities?”
 - “The Gap-closing Estimand” (Lundberg 2024) or “Disparity Treatment Effect” (Shiba)
 - Example: Social isolation and social inequalities in all-cause mortality (Lunar et al. 2025)

Summary 1

- Two key ingredients of health disparities
 - Effect heterogeneity
 - Exposure disparities
 - Often ignored by considering the “all exposed vs unexposed” scenarios
- Standardization to predict potential outcome distributions under hypothetical stochastic interventions
 - Consider estimand that explicitly captures changes in disparities
 - Disparity effect

Mediation Analysis

Two effects in mediation analyses



- **Indirect Effect**

→ Effect of exposure A through mediator M

- **Direct effect**

→ Effect of exposure A that is independent of mediator M

- Various definitions depending on the motivation
- Should be chosen based on causal estimand of interest.

Motivation for Causal Mediation Analysis

- **#1 Target of interventions**

→ Particularly useful when “exposure” is hard to manipulate

- **#2 Effect decomposition**

→ Explaining what explains the effect that we are observing

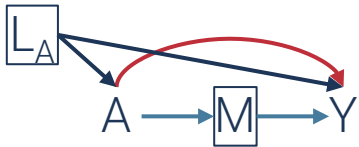
→ Strengthening theory and generating hypotheses

Naive Approach: Difference Method

Example: Linear regression of outcome Y on exposure A adjusting for confounders L_A



- Model 1: $E[Y|A, L_A] = \beta_0 + \beta_1 A + \beta L_A$
 - β_1 : Total effect of A on Y (within strata of L_A)
 - Assuming exchangeability conditional on L_A

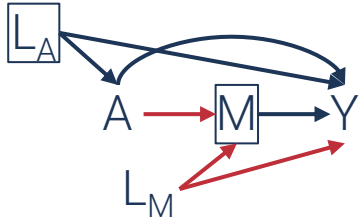


- Model 2: $E[Y|A, M, L_A] = \gamma_0 + \gamma_1 A + \gamma_2 M + \gamma L_A$
 - Additionally adjust for mediator M
 - Attenuation as “evidence of mediation”
 - γ_1 : Effect of A conditional on M
 - Effect that is independent of M
 - Direct effect
 - $\beta_1 - \gamma_1$: Effect of A through M
 - Indirect Effect

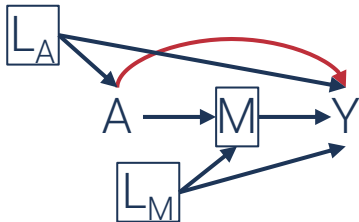
⚠ Any potential issue?

Naive Approach: Difference Method

Example: Linear regression of outcome Y on exposure A adjusting for confounders L_A



- Model 2: $E[Y|A, M, L_A] = \gamma_0 + \gamma_1 A + \gamma_2 M + \gamma L_A$
 - γ_1 : Association between A and Y conditional on M
 - M as a collider
 - Selection bias through mediator-outcome confounders L_M
 - Ignored exposure-mediator interaction $A \times M$



- Model 3:
 $E[Y|A, M, L_A, L_M] = \theta_0 + \theta_1 A + \theta_2 M + \theta_3 A \times M + \theta L_A + \delta L_M$
 - $\theta_1 : E[Y|A = 1, M = 0, L_A, L_M] - E[Y|A = 0, M = 0, L_A, L_M]$

💡 Corresponding causal estimand?

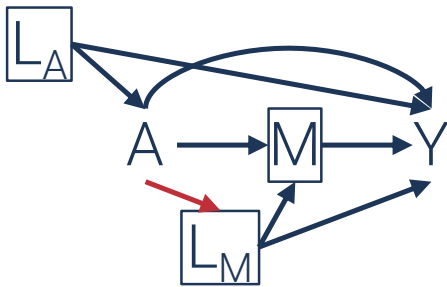
- $E[Y^{a=1, m=0} - Y^{a=0, m=0} | L_A, L_M]$ under conditional exchangeability for BOTH A and M
 - “The effect of A while fixing the mediator to $m = 0$ ”
 - Joint intervention for both A and M
 - Controlled Direct Effect

Causal Mediation Analysis

- Naive regression analysis is often:
 - Based on strong and potentially incorrect assumptions
 - Vague or potentially irrelevant causal estimand
- **Causal mediation analysis**
 - First, **define Direct/Indirect Effects based on potential outcomes**
 - CDE, NDE, IDE, etc
 - Second, consider identification assumptions
 - Third, estimate the target statistical estimand
 - Can be regular outcome regression
 - if assumptions are valid and corresponding estimand is of interest
 - But also flexible estimation via weighting, machine learning, etc

Controlled Direct Effect

- **Total Effect:** $TE = E[Y^{a=1} - Y^{a=0}]$
- **Controlled Direct Effect:** $CDE(m) = E[Y^{a=1,m} - Y^{a=0,m}]$
 - Function of a specific mediator value m
- “Proportion eliminated”: $(TE - CDE(m))/TE$
 - How much of the total effect of A would be eliminated if we intervened and fixed mediator value to m for everyone?



- CDE as a joint intervention for A and M
 - Can be viewed as time-varying treatment analysis
 - CDE can be identified in the presence of **exposure-induced confounding**: $A \rightarrow L_M$
 - Adjusting for time-dependent confounding via g-methods
 - e.g., “Causal mediation via marginal structural model” (Nandi et al. 2012)

“Hypothetical Mediator Intervention”

- Controlled Direct Effect: $CDE(m) = E[Y^{a=1,m} - Y^{a=0,m}]$
 - Joint intervention that also fixes mediator to a specific value m
 - Deterministic intervention for M
 - Different $CDE(m)$ estimates across m values with the exposure-mediator interaction
 - How do we choose a meaningful m value?
 - Policy relevance (might be feasible with a binary M)
 - But hard to pick a single value for M with numerous levels
 - e.g., fixing PM2.5 to $9\mu g/m^3$ for everyone

Alternative hypothetical mediator intervention

- A random draw from a distribution of mediator values $\mathcal{M}$
 - e.g., $\mathcal{M}(a|L)$
 - “Distribution of mediator values that would have been observed among people like you (in terms of covariates L) had they received treatment a ”

“Hypothetical Mediator Intervention” for a Mediator Distribution

- **Interventional Direct Effect**

→ $IDE = E[Y^{a=1, \mathcal{M}(0|L)} - Y^{a=0, \mathcal{M}(0|L)}]$

- Exposure effect that would be observed while fixing the *distribution* of mediator values
 - Distribution under $a = 0$ conditional on L
 - Additionally require no A-M confounding assumption
- Can also be defined as $IDE = E[Y^{a=1, \mathcal{M}(1|L)} - Y^{a=0, \mathcal{M}(1|L)}]$

→ Non-zero IDE

- Suggests the presence of effect of A that is independent of $\mathcal{M}$
 - Hence, direct effect

- Extending causal inference for distributional interventions (see Lecture 5) to mediation

→ Stochastic interventional effect (Rudolph et al. 2018)

- aka interventional effects, randomized interventional analogs

→ Flexible specification of $\mathcal{M}$

Extension to Health Disparity Research



- Example in Nguyen et al. (2021)

→ Y: Wellbeing
 → M: Bullying
 → A: Sexual minority

- Observed disparity:

→ $\mathcal{D} = E[Y|A = 1] - E[Y|A = 0]$

- Counterfactual disparity under a hypothetical intervention (A^g) equalizing the level of bullying $\mathcal{M}$ among sexual minority to the level experienced by majority (i.e., $\mathcal{M}(0)$)

→ $\mathcal{D}^{A^g} = E[Y^{\mathcal{M}(0)}|A = 1] - E[Y|A = 0]$

- **“Decomposing” disparity**

→ $\mathcal{D} = E[Y|A = 1] - E[Y|A = 0]$

□ $= E[Y|A = 1] - E[Y^{\mathcal{M}(0)}|A = 1] + E[Y^{\mathcal{M}(0)}|A = 1] - E[Y|A = 0]$

□ $= E[Y|A = 1] - E[Y^{\mathcal{M}(0)}|A = 1] + \mathcal{D}^{A^g}$

Nguyen et al. (2021) used the notation $E[Y^{1, \mathcal{M}(0, C)}|A = 1, C] - E[Y|A = 0, C]$ for $\mathcal{D}^{A^g}$, which I think is a typo because they conceptualized no hypothetical intervention for sexual minority status. Also note that their estimation is conditional on covariates C . A marginal rather than conditional distribution $\mathcal{M}$ may

Extension to Health Disparity Research

- “Decomposing” disparity
 - $\mathcal{D} = E[Y|A = 1] - E[Y^{\mathcal{M}(0)}|A = 1] + \mathcal{D}^{A^g}$
 - Observed disparity = “Disparity-removed” + Remaining disparity under an intervention
- Causal inference for health disparity
 - Often viewed as an extension of effect decomposition/mediation
 - But the causal estimand involves no counterfactual for the “exposure”
 - e.g., no conceptualization of “race effect”
 - Avoid the potential consistency concern for the “exposure”
 - No components corresponding to “direct/indirect effects”
 - (My view) a more general extension of stochastic intervention effect rather than mediation

! Which is more relevant?

- The decomposition?
- The expected change in disparity (i.e., $\mathcal{D} - \mathcal{D}^{A^g}$)?
 - The “Gap-closing Estimand” (Lundberg 2024) or “Disparity (Treatment) Effect” (Shiba)

Disparity Effect and Effect Decomposition

- Example from [this slide](#)
- “Disparity effect”
 - $(E[Y^{A^g}|SES = high] - E[Y^{A^g}|SES = low]) - (E[Y|SES = high] - E[Y|SES = low])$
 - “How would the hypothetical intervention change SES-based health disparities?”
 - e.g., what if the distribution of A among $SES = low$ is identical to that of $SES = high$
 - Can conceptualize alternative interventions more flexibly
 - Simulating how interventions may change “exposure disparities” ([see this slide](#))

Different notation/estimand but same idea

- Mediation would have conceptualized SES as an “exposure”
 - Here, the “exposure” is what mediation would conceptualize as a mediator (M)
 - Correspond better to conceptualized hypothetical interventions (A^g)
 - Manipulating the distribution of A while conducting no explicit intervention for SES
- More explicit focus on consequence rather than explanation
- Can incorporate “allowability” ([Jackson 2021](#))

Disparity Effect and Effect Decomposition

- Framing & terminology do not matter as long as you are:
 - Explicit about causal estimand of interest
 - Explicit about a hypothetical intervention under consideration
 - Social identity as a factor that defines disparity rather than an interventional target
- Many estimators
 - Standardization
 - Explicit focus on the two key ingredients of disparity
 - Effect heterogeneity & exposure disparities
 - Potentially different insights depending on which is the primary driver
 - IPW
 - Doubly-robust
 - Modeling of the ingredients tended to be avoided or treated implicitly as intermediate estimation steps

Disparity Effect and Effect Decomposition

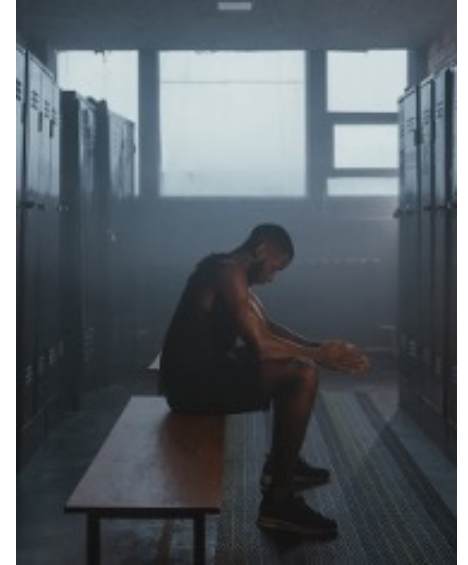
Further readings

- **On the importance of heterogeneity in disparity research**
 - Ward et al. (2019) & Shiba and Inoue (2024)
- **Decomposition/mediation framework**
 - Jackson and VanderWeele (2018), Jackson (2021), Jackson and VanderWeele (2019)
 - Comprehensive summary by Qin and Jackson (2025) & Chang et al. (2024)
 - On using CDE in disparity research by Naimi et al. (2016)
 - Non-parametric approach by Yu and Elwert (2025)
- **More general disparity estimand**
 - Excellent review by Opacic et al. (2025)
 - “Gap-closing estimand” by Lundberg (2024)
 - Recent development by Jackson et al. (2024)
 - Wang et al (forthcoming)

Simulating Disparity Effects: An Applied Example

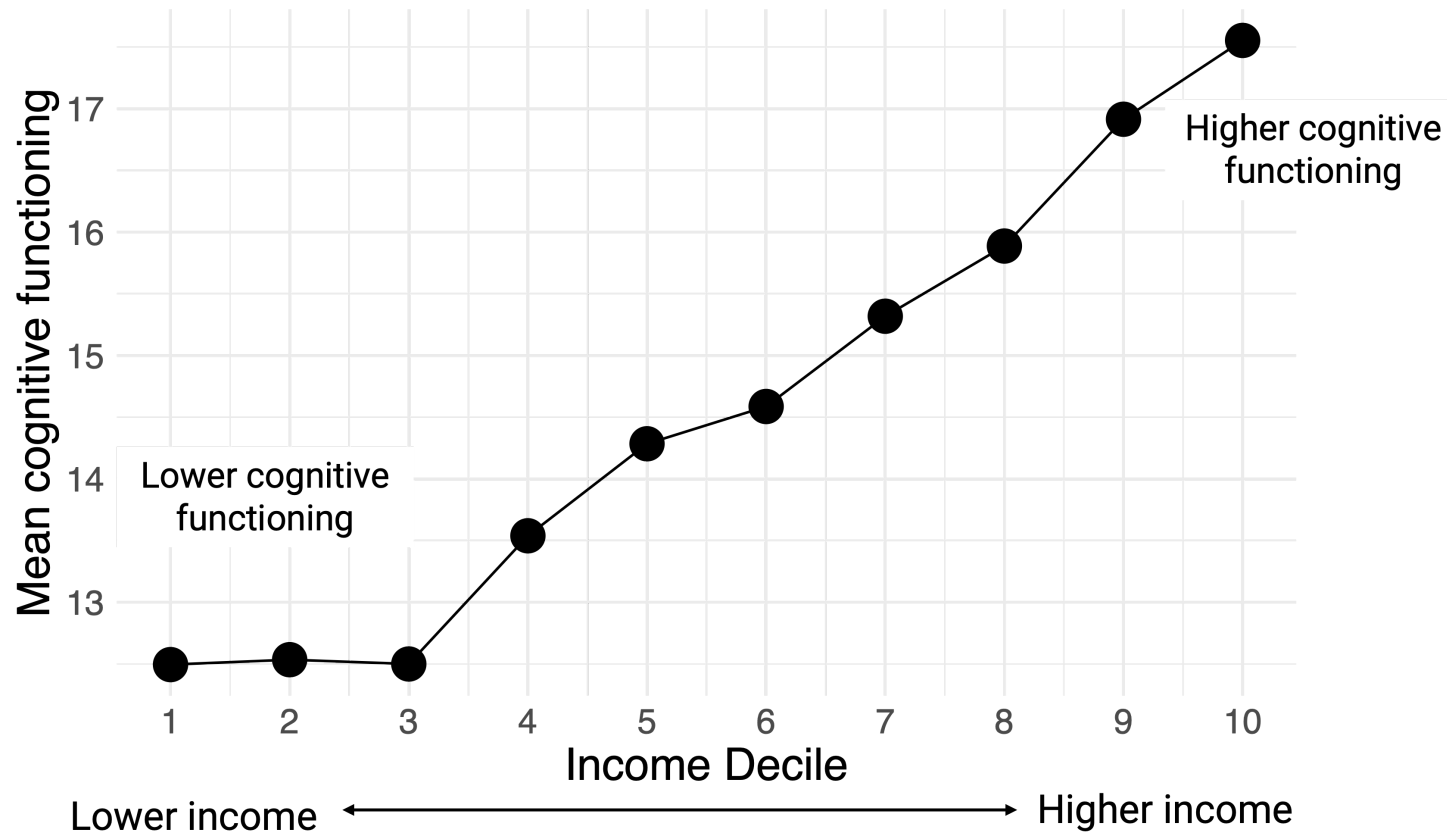
Social isolation and cognitive functioning of older adults (from Lecture 6)

- **Data:** Health and Retirement Study (n=11,322)
- **Exposure:** Social isolation (dichotomized composite index)
 - Marital status and cohabitation
 - Contact with children, family, and friends
 - Social participation
- **Outcome:** Cognitive functioning (TICS; 0-27 points)
- **Covariates:** 35 pre-baseline covariates
- **Analysis:**
 - Non-parametric estimation of CATE ($E[Y^{a=1} - Y^{a=0}|L]$) via Causal Forest
 - **Simulated changes in disparities in cognitive functioning** using the standardization approach described in this slide



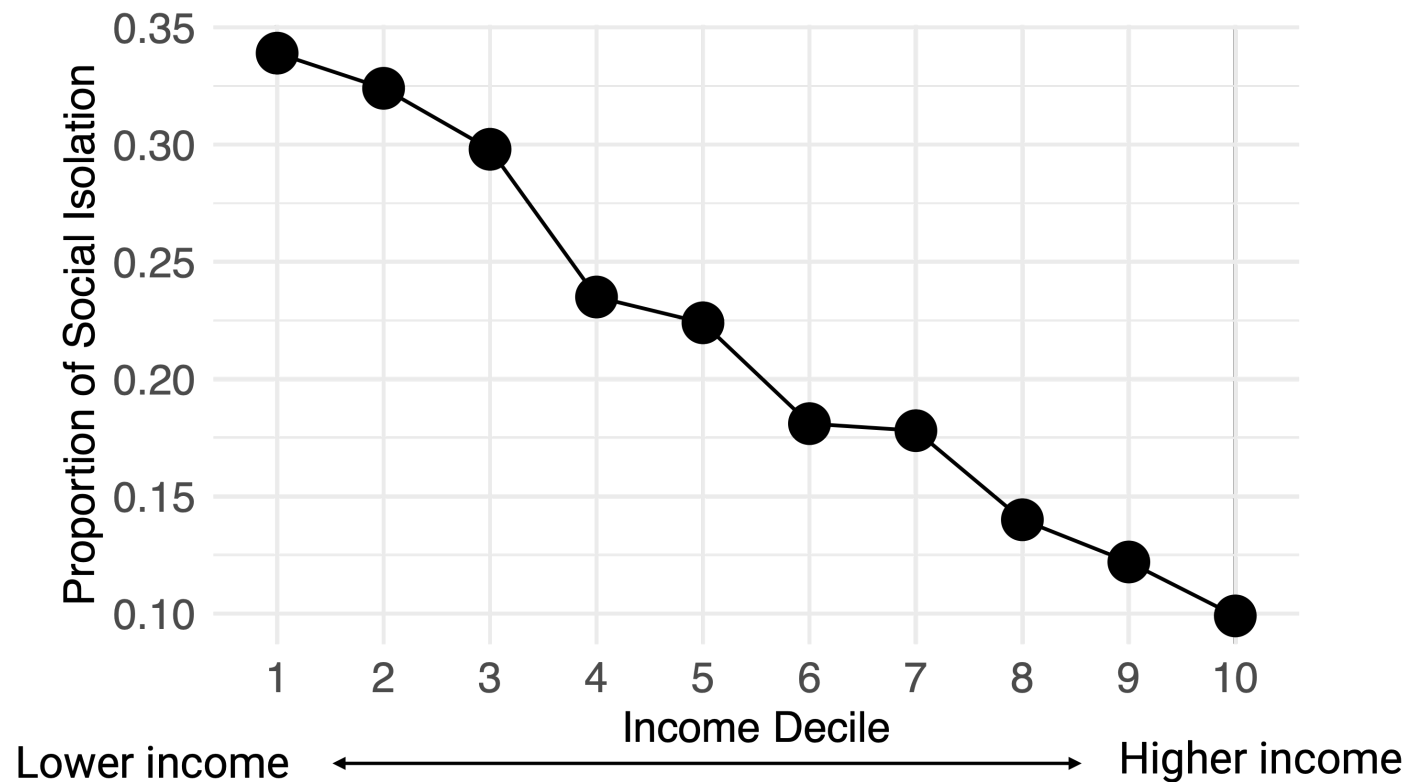
Observed Outcome Disparities

- Lower income was associated with lower levels of cognitive functioning



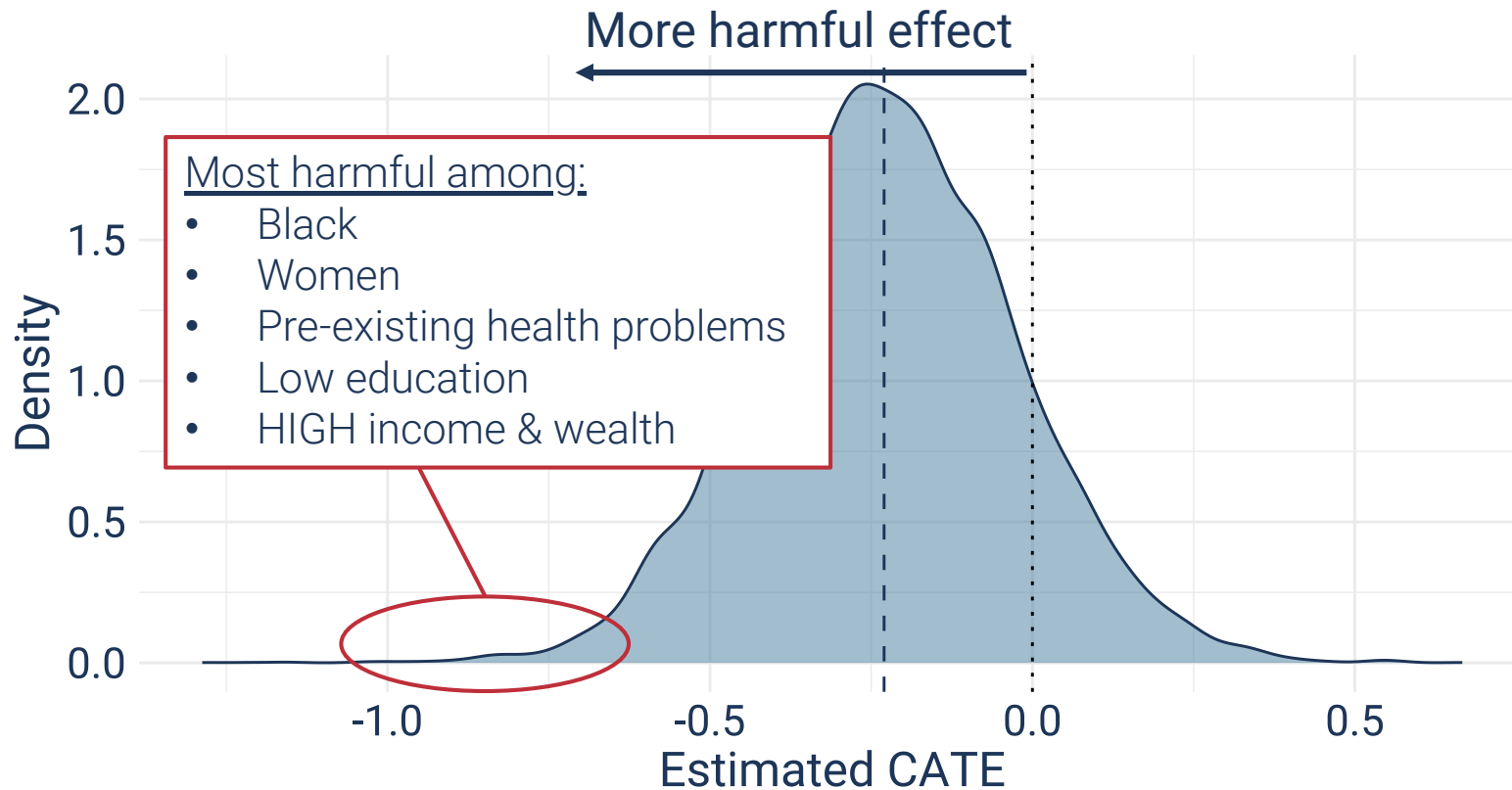
Ingredient 1: Exposure Disparity

- Lower income was associated with higher prevalences of social isolation
 - Same trends for wealth, education, and race and ethnicity
 - Socially marginalized groups are more likely to be *exposed* to social isolation



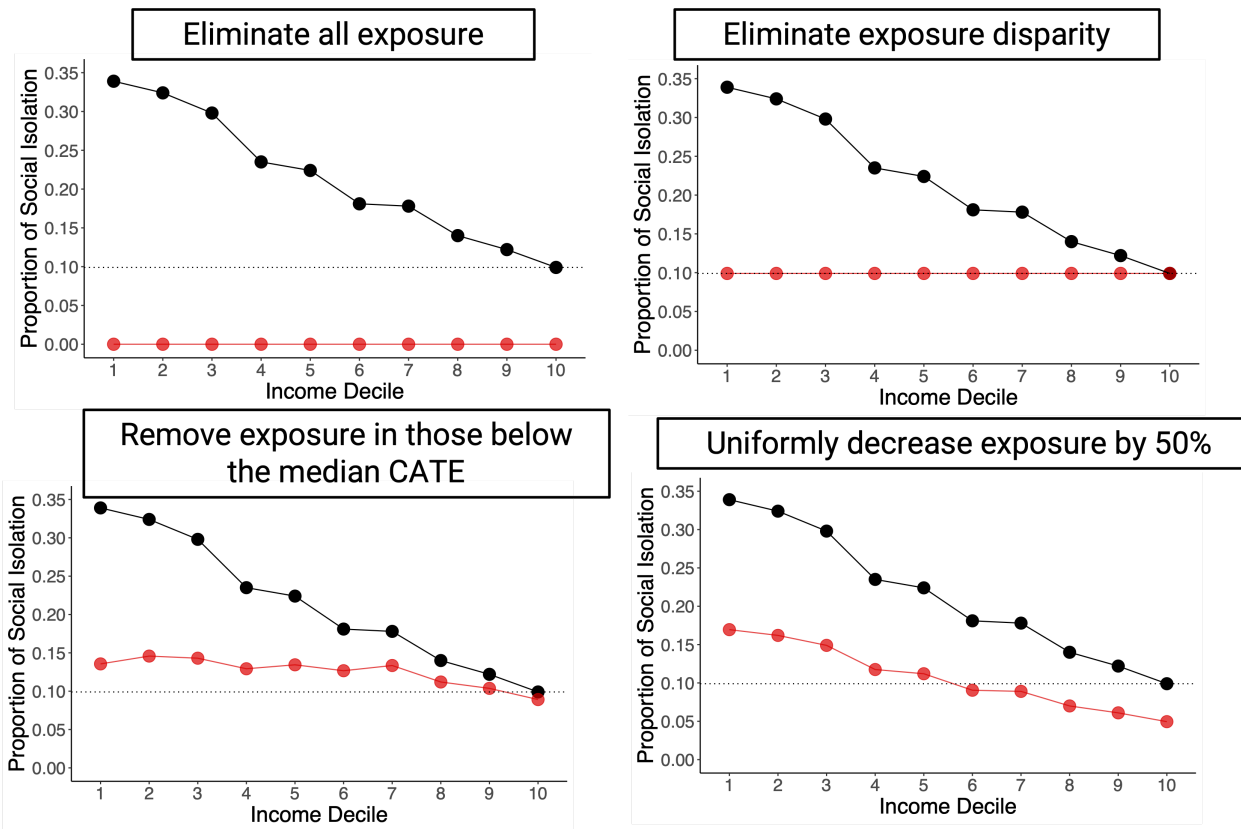
Ingredient 2: Effect Heterogeneity

- Socially marginalized groups are *harm*ed more severely by social isolation
 - Despite the notable trend by intersections of high income and other forms of social marginalization (small in size)



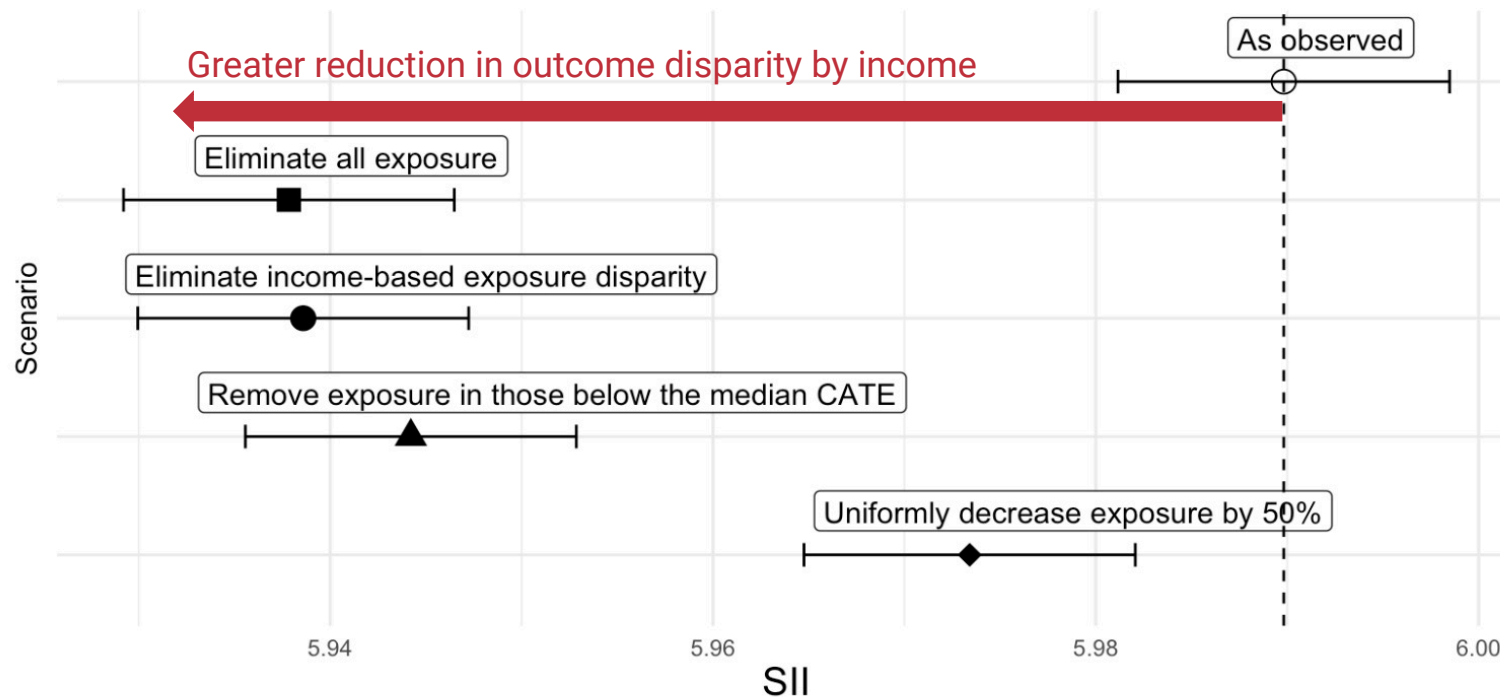
Hypothetical Interventions

- Interventions that remove social isolation among the exposed
 - Explicitly addressing exposure disparities
 - Preserve exposure disparities while reducing the population-wide prevalence



Simulated Disparity Effects

- Population interventions for social isolation would reduce income-based disparities in cognitive functioning
 - Same trend for wealth, education, and race and ethnicity
 - Particularly so when the intervention explicitly addresses exposure disparities



Summary 2

- **Controlled Direct Effect (CDE)**

- Joint intervention for both exposure and mediator
 - Fixing mediator to a specific value
- Conditional exchangeability assumptions
 - No A-Y confounding, no M-Y confounding

- **Interventional Direct Effect (IDE)**

- Joint intervention for both exposure and mediator
 - Setting a mediator distribution
- Conditional exchangeability assumptions
 - Same as CDE + no A-M confounding

- **Disparity Effect**

- Extension of stochastic intervention analysis
- Can be also viewed as “effect” decomposition
 - But no causal “effect” conceptualized for exposure

Mediation for Effect Decomposition

- CDE as a joint intervention

- “What is the remaining effect of A if I fix M to m via an intervention?”

- Does not capture the extent to which mediator *explains* the exposure effect that they would experience without the mediator intervention
 - ... because people may take various mediator values under the intervention $A \rightarrow a$
 - Counterfactual mediator values M^a
 - Mediator level that a *person* would have *naturally* experienced had the *person* received a

- **Natural Direct Effect**

- $NDE = E[Y^{a=1, M^1}] - E[Y^{a=0, M^1}]$

- Can be also defined as $E[Y^{a=1, M^0}] - E[Y^{a=0, M^0}]$

- What is the remaining effect of A if mediator is set to the same level for each individual?

- Hence, direct effect

Identification for Natural Effects

- Conceptualizing the potential outcome $Y^{a,M^{a*}}$
 - Involves the effect of A on M
 - Another set of identification assumptions
 - Conditional exchangeability, consistency, and positivity for the A-M relationship
 - Two different counterfactuals embedded simultaneously
 - “Cross-world independence”
 - Additional assumption: No exposure-induced M-Y confounding

NDE/NIE requires stronger assumptions than CDE

Mediation for Effect Decomposition

- Total effect $TE = E[Y^{a=1}] - E[Y^{a=0}]$
 - $TE = E[Y^{a=1,M^1}] - E[Y^{a=0,M^0}]$ by consistency for the effect of A on M
 - $E[Y^{a=1,M^1}] - E[Y^{a=0,M^1}] + E[Y^{a=0,M^1}] - E[Y^{a=0,M^0}]$
 - $NDE + NIE$
- **“Natural Indirect Effect (NIE)”**
 - The magnitude of exposure effect mediated through M
 - “Proportion mediated”: NIE/TE
 - Distinguish this from “proportion eliminated” in [this slide](#)
 - Read VanderWeele (2013) on their differences
- Effect can be decomposed into smaller components
 - e.g., separating effects attributable to effect heterogeneity or mediation
 - Can be useful to test alternative hypotheses about mechanisms
 - e.g., life course epidemiology (Yazawa et al. 2022)

Summary 3

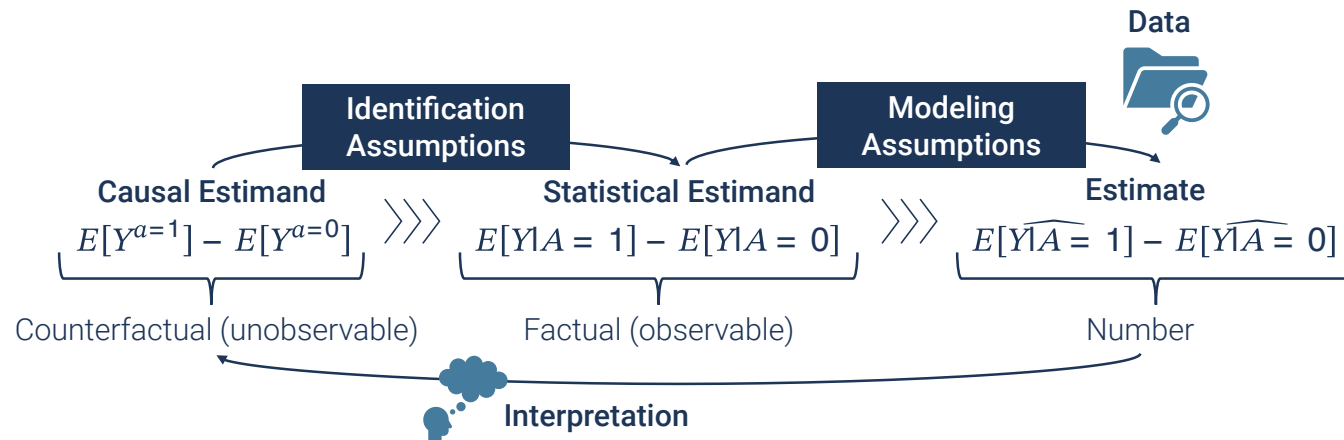
- **Natural Direct Effect (NDE)**

- Definition based on cross-world counterfactuals
- Useful for effect decomposition
 - Understanding mechanisms
- Conditional exchangeability assumptions
 - No A-Y confounding, no A-M confounding, no M-Y confounding affected by A (i.e., no exposure-induced confounding)

- **CDE vs NDE vs IDE, etc**

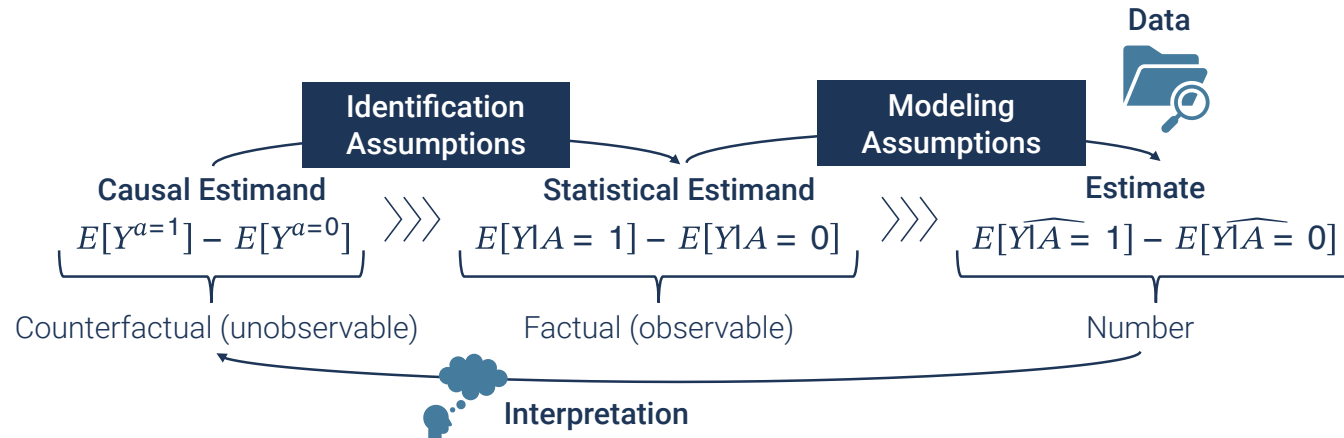
- Know differences in definitions (estimand), identification assumptions, implications
- Helpful tutorials: VanderWeele (2016) & Nguyen et al. (2021) & Nguyen et al. (2022)

How to Critically Assess Methodology in (Social) Epi Papers



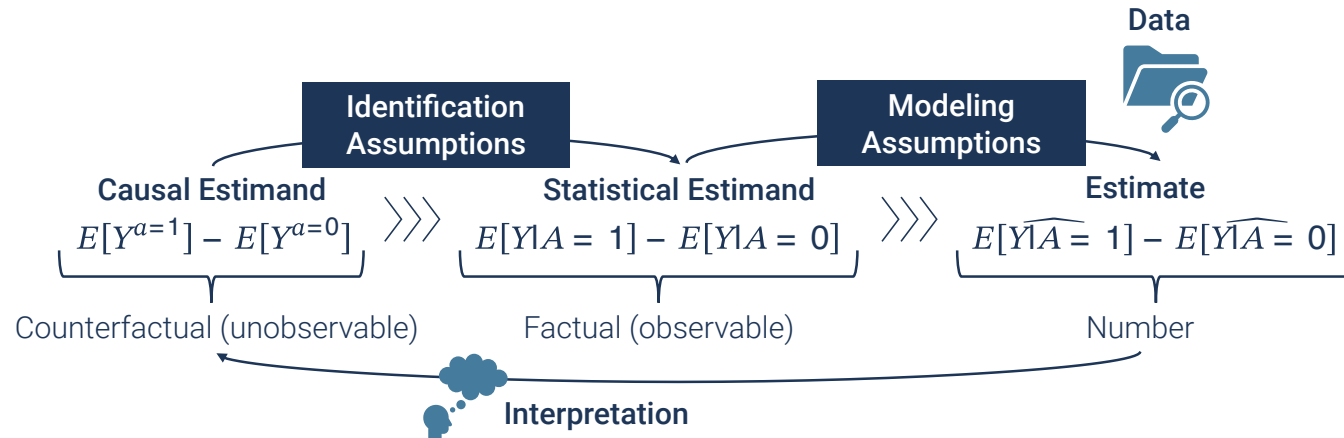
- **#1. Identify causal question under consideration**
 - Often vague or not useful
 - e.g., not specifying time interval
 - **Is it well-justified?**
 - e.g., Using g-methods when a single-point intervention is sufficient
- **#2. Check correspondence between what they examined vs what they claim they did**
 - Discrepancy may arise from:
 - Statistical approach (e.g., using IV without claiming complier effects)
 - Measurement with a conceptual validity issue

How to Critically Assess Methodology in (Social) Epi Papers



- **#3. Assess the validity of underlying assumptions**
 - Causal identifications
 - Conditional exchangeability & consistency
 - Modeling assumptions (e.g., linearity)
- **#4 Check correspondence between numeric estimates and their interpretation**
 - e.g., interpreting effect heterogeneity as causal interaction
 - Did they discuss alternative explanations? (e.g., multiple versions of treatment)
 - e.g., no association with baseline exposure as evidence of no effect at all
 - absence of evidence is not evidence of absence
 - but also potential non-adherence (i.e., time-varying treatment)

When You Learn Methods



- **It is all about trade-off**

- No single “winning” method
- Fancy methods $\neq$ better methods
- Consider estimand and assumptions
 - More relevant estimand & stronger assumptions vs vague estimand & weaker assumptions

- **Distinguish identification and estimation**

- Which of these two is the method improving?
- e.g., ~~“Propensity score matching can estimate causal effects”~~
- e.g., ~~“Machine learning methods must not be used in causal inference”~~

When You Learn Methods

- **Focus on mathematically defined causal estimand**
 - Try not to be obsessed with terminology
 - e.g., effect modification, effect heterogeneity, causal estimand
 - Often overlapping concepts and methods but different terminology
 - e.g., Controlled direct effect as a time-varying treatment effect
 - e.g., Causal interaction as a joint intervention
 - Can use the time-varying treatment analysis when conceptualizing a joint intervention for temporally separated targets
 - e.g., disparity effect vs “effect” decomposition/mediation

⚠ Always keep in mind

- Many different approaches to the same problem
- There is no single “correct” view/approach
 - Be flexible and creative

Announcements

- Final Assignment (due October 24th at 11:59 PM)
- Course survey
 - Formal and informal surveys
- Photo & Marketing
- Research projects

References

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